

CANONICAL LESSON

Anomaly Is Not Guilt

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C46
Notebook	V7_C_N11
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Use examination anomalies as review signals—not proof of misconduct—while preserving evidence, uncertainty, confidentiality, due process, and authorized investigative authority.

Learning outcomes

Why this matters

Automated anomaly detection can wrongly accuse learners or schools when similarity, context, data quality, and alternative explanations are ignored.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

An assessment authority examines unusual answer patterns with administration records and independent review before drawing any conclusion.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Control multiple testing.
- 2.
- Explain similarity limits.
- 3.

- Design due process.

Answer key

- 1.
- Pre-specify tests and use false-discovery or family-wise controls with validation.
- 2.
- Similarity can reflect teaching, curriculum, chance, form structure, or accommodations; it is not proof of collusion.
- 3.
- Restrict access, preserve evidence, notify fairly, allow response/appeal, require independent human review, and separate analytics from sanction authority.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.